Patient Name : EDRU RAMESHKUMAR
Age / Gender : 62 years / Male
Patient ID : 12527
Source : DIRECT

Referral : Dr. MADHAVA RAO
Collection Time : May 20, 2022, 06:17 a.m.
Receiving Time : May 20, 2022, 06:26 a.m.
Reporting Time : May 20, 2022, 08:34 a.m.
Sample ID : 0221400003

Test Description	Result(s)	Biological Reference Intervals	Unit(s)
Thyroid Function Test			
*T3	105.22	87 - 178	ng/dL
*T4	8.1	4.82 - 15.65	µg/dL
*TSH	1.60	0.38 - 5.33	µIU/mL

Method: Chemi Luminescence Immunoassay(Sample Type:Serum)

Interpretation

- Measurement of TSH in conjunction with T3,T4 or FT3,FT4 is used for assessment of thyroid gland functional status and also for monitoring patients on treatment for hyperthyroidism & hypothyroidism. Suppressed TSH (<0.01 µIU/mL) suggests a diagnosis of hyperthyroidism and elevated concentration (>7 µIU/mL) suggest hypothyroidism. TSH levels may be affected by acute illness and several medications including dopamine and glucocorticoids.TSH concentration are decreased (low or undetectable) in Graves disease. Increased in TSH secreting pituitary adenoma (secondary hyperthyroidism); PRTH. TSH concentration are primary elevated in hypothyroidism (along with decreased T4) except for pituitary & hypothalamic disease.
- Mild to modest TSH elevations in patient with normal T3 & T4 levels indicates impaired thyroid hormone reserves & incipent hypothyroidism (subclinical hypothyroidism).
- Mild to modest decrease in TSH with normal T3 & T4 indicates subclinical hyperthyroidism.
- Degree of TSH suppression does not reflect the severity of hyperthyroidism, therefore, measurement of free thyroid hormone levels is required in patient with a supressed TSH level.
- If free T4 is normal, free T3 should be checked as it is the first hormone to increase in early hyperthyroidism.Though TSH levels can also be used to effectively monitor patients being treated with thyroid hormones but it may be misleading. Therefore total or free T4 generally serve as front line assay during this period. The free T3 & T4 (FT3 & FT4) measures concentrations of free hormones, which are not affected by changing in concentrations of binding proteins, therefore more reliable indicator of true thyroid status.

VITAMIN B12			
***Vitamin B12** CLIA **Method : CLIA ChemiLuminescence Immunoassay**	498	222-1439	pg/mL

Method: ChemiLuminescence Immunoassay(Serum)

Interpretation:

1. Vit B12 levels are decreased in megaloblastic anemia, partial/total gastrectomy, pernicious anemia, peripheral neuropathies, chronic alcoholism, senile dementia, and treated epilepsy.
2. An associated increase in homocysteine levels is an independent risk marker for cardiovascular disease and deep vein thrombosis.
3. HoloTranscobalamin II levels are a more accurate marker of active VitB12 component.

Note: Patients on Biotin supplement may have interference in some immunoassays. With individuals taking high dose Biotin (morethan 5 mg per day) supplements, at least 8-hour wait time before blood draw is recommended.
Ref: Arch Pathol Lab Med—Vol 141, November 2017

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Test Description	Result(s)	Biological Reference Intervals	Unit(s)
Note			

END OF REPORT

Approved By

Dr. V. Madhava Rao
MD (PGI-Chandigarh)
DM (Endocrinology) (AIIMS-New Delhi)

Patient Name : EDRU RAMESHKUMAR	**Referral :** Dr. MADHAVA RAO
Age / Gender : 62 years / Male	**Collection Time :** May 20, 2022, 06:17 a.m.
Patient ID : 12527	**Receiving Time :** May 20, 2022, 06:26 a.m.
Source : DIRECT	**Reporting Time :** May 20, 2022, 09:08 a.m.
	Sample ID : 0221400003

Test Description	Result(s)	Biological Reference Intervals	Unit(s)
FASTING PLASMA GLUCOSE			
Fasting Plasma Glucose (F) Method : Glucose oxidase/Peroxidase	97.73	Desired Fasting Plasma Glucose: For nondiabetic: 70 - 100 mg/dL For Diabetic: 80 - 130 mg/dL	mg/dL
.			
POST PRANDIAL PLASMA GLUCOSE			
Post Prandial Plasma Glucose Method : Glucose oxidase/Peroxidase	114.22	< 180	mg/dL

Instrument:Done on BIOSYSTEMS BA 400, a Fully Automated Biochemical Analyser

Note

Test Description	Result(s)	Biological Reference Intervals	Unit(s)
HbA1c			
HbA1c **Method:HPLC (BIORAD D 10)(Sample Type:Blood)**	5.8	Non Diabetic : 4.0 - 5.6 % Border line : 5.7 - 6.4 % Good control in Diabetes : < 6 Years : < 8.5 % 6 Years - 12 Years : < 8.0 % 13 Years - 18 Years : < 7.5 % > 18 Years : < 7 %	%
Estimated Average Glucose (eAG) Calculated	119.76		mg/dL

Interpretation :

1. HbA1c is used for monitoring diabetic control. It reflects the estimated average glucose (eAG).
2. HbA1c has been endorsed by clinical groups & ADA (American Diabetes Association) guidelines 2012, for diagnosis of diabetes using a cut-off point of 6.5%. ADA defined biological reference range for HbA1c is 4% to 5.7%. Patients with HbA1c value between 5.7% to 6.5% are considered Pre-diabetic.
3. Trends in HbA1c are a better indicator of diabetic control than a solitary test.
4. Low glycated haemoglobin(below 4%) in a non-diabetic individual are often associated with systemic inflammatory diseases,chronic anaemia(especially severe iron deficiency &haemolytic), chronic renal failure and liver diseases. Clinical correlation suggested.
5. To estimate the eAG from the HbA1C value, the following equation is used: eAG(mg/dl) = 28.7*A1c-46.7
6. Interference of Haemoglobinopathies in HbA1c estimation.

A. For HbF > 25%, an alternate platform (Fructosamine) is recommended for testing of HbA1c.
B. Homozygous hemoglobinopathy is detected, fructosamine is recommended for monitoring diabetic status
C. Heterozygous state detected is corrected for HbS and HbC trait.

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Test Description	Result(s)	Biological Reference Intervals	Unit(s)

Hemoglobin electrophoresis (HPLC method) is recommended for detecting hemoglobinopathy.

Note

CREATININE

Test Description	Result(s)	Biological Reference Intervals	Unit(s)
CREATININE Method:JAFFE COMPENSATED (Sample Type:Serum)	0.93	0.7 - 1.2	mg/dL

Instrument:Test Done On Fully Automated Biochemistry Analyser, BIOSYSTEMS BA 400

Note

UREA

Test Description	Result(s)	Biological Reference Intervals	Unit(s)
UREA Method:UREASE / GLUTAMATE DEHYDROGENASE(Sample Type:Serum)	20.24	Cord: 21 - 40 mg/dL Pre - Mature : 3 - 20 mg/dL New Born : 4 - 12 mg/dL Child : 5 - 18 mg/dL Adult : 15 - 39 mg/dL <60 Years : 8 - 23 mg/dL	mg/dL

Note

S.IRON

Test Description	Result(s)	Biological Reference Intervals	Unit(s)
IRON Method:FERROZINE(Sample Type:Serum)	92.92	Men : 65 - 175 µg/dL Women : 50 - 170 µg/dL	µg/dL

Instrument:Test Done On Fully Automated Biochemistry Analyser, BIOSYSTEMS BA 400

Note

LIPID PROFILE

Test Description	Result(s)	Biological Reference Intervals	Unit(s)
TOTAL CHOLESTEROL Method:Cholestrol Oxidase/Peroxidase(Sample Type:Serum)	125.78	Desirable : Up to 200 mg/dL Borderline High: 200-239 mg/dL High : > 240 mg/dL	mg/dL
TRIGLYCERIDES Method:Glycerol phosphate oxidase/Peroxidase(Sample Type:Serum)	**179.19**	Normal : Up to 150 mg/dL Borderline High : 150-199 mg/dL High : 200-499 mg/dL Very high : > 500 mg/dL	mg/dL
HDL CHOLESTEROL Method:Direct(Sample Type:Serum)	39.23	> 40 mg/dL High Risk : < 35 mg/dL Low Risk : > 60 mg/dL	mg/dL

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Test Description	Result(s)	Biological Reference Intervals	Unit(s)
*LDL CHOLESTEROL Calculated	50.71	Optimal : Up to 100 mg/dL Near optimal/above optimal : 100-129 mg/dL Borderline High : 130-159 mg/dL High : 160-189 mg/dL Very High : > 190 mg/dL	mg/dL
*VLDL CHOLESTEROL Calculated	35.84	5 - 51 mg/dL	mg/dL
*CHOL / HDL RATIO Calculated	3.21	Low risk : < 3.5 Average risk : 3.5 - 5.0 High risk : > 5.0	
*LDL/HDL RATIO Calculated	1.29	Desirable : < 0.5-3.0 Borderline : 3.1- 6.0 Elevated : > 6.0	

INSTRUMENT: DONE ON BA 400 BIOSYSTEMS FULLY AUTOMATED BIOCHEMISTRY ANALYZER

Note

Liver Function Tests

Test Description	Result(s)	Biological Reference Intervals	Unit(s)
BILIRUBIN (TOTAL) DICHLOROPHENYL DIAZONIUM (Sample Type:Serum)	0.56	<24Hrs: 1.0 - 8.0 mg/dL <48Hrs: 6.0 - 12.0 mg/dL 3 - 5 Days: 10.0 - 14.0 mg/dL Adults : <1.0 mg/dL	mg/dL
*BILIRUBIN (DIRECT) DICHLOROPHENYL DIAZONIUM (Sample Type:Serum)	**0.33**	< 0.2 mg/dL	mg/dL
BILIRUBIN (INDIRECT)(Sample Type:Serum) calculated Method : Calculated	0.23	0.2 -0.8 mg/dL	mg/dL
SGOT IFCC without pyridoxal 5-P(Sample Type:Serum)	14.48	0 - 40	U/L
SGPT IFCC without pyridoxal 5-P(Sample Type:Serum)	16.11	0 - 41	U/L
TOTAL PROTEIN BIURET (Sample Type:Serum)	7.12	Pre mature : 0.6 -6.0 g/dL New born : 4.6 - 7.0 g/dL One week : 4.4 - 7.6 g/dL 7 months - 1 year : 5.1-7.3 g/dL 1 Year - 2 years : 5.6 - 7.5 g/dL > 3 years : 6.5 - 8.0 g/dL	g/dL
ALBUMIN BROMOCRESOL GREEN(Sample Type:Serum)	4.22	3.4-4.8 g/dL	g/dL

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Test Description	Result(s)	Biological Reference Intervals	Unit(s)
*GLOBULIN	2.90	2.5 - 4.5	g/dL
Calculated			
*A:G RATIO	**1.46**	1.5 - 3.5	
ALP (ALKALINE PHOSPHATASE)	93.17	<115 U/L	U/L
2Amino 2Methyl - 1Propanol Buffer IFCC Method(Sample Type:Seru			
*gGT	23.76	< 55	U/L
Method : IFCC			

S.PHOSPHORUS

PHOSPHORUS	2.96	2.5 - 4.5	mg/dL
Method: PHOSPHOMOLYBDATE/UV (Sample Type:Serum)			

Method: TEST DONE ON BIOSYSTEMS BA 400. A FULLY AUTOMATED BIOCHEMISTRY ANALYSER

HEMOGRAM (CBP)

*Haemoglobin	14.1	12.0 - 18.0	g/dL
*R.B.C count	4.98	3.80 - 5.30	10^6µL
*Leucocyte Count (WBC Count)	8200	4000 - 11000	µL
*Platelet Count	212000	120000 - 380000	µL
*PCV	43.9	36.0 - 56.0	%
*MCV	88.2	80.0 - 100.0	fL
*MCH	28.3	27-32	pg
*MCHC	32.1	32.0 - 36.0	g/dL
*RDW-CV	13.3	10.0 - 16.5	%
*Mean Platelet Volume (MPV)	6.3	5.0 - 10.0	fL
*PDW	17.8	12.0 - 18.0	%

***DIFFERENTIAL COUNT**

*Granulocyte	67.6	42 - 85	%
*Lymphocytes	28.5	11.0 - 49.0	%
*Monocytes	3.9	0.0 - 9.0	%

Instrument: TEST PERFORMED ON NIHON KOHDEN HEMATOLOGY ANALYZER

Note

COMPLETE URINE EXAMINATION

Colour	Pale yellow	Pale yellow	
(Sample Type:Urine)			
pH	5.0	4.5 – 8.5	
Specific Gravity	1.020	1.015 – 1.025	
Glucose	Negative	Negative	

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Test Description	Result(s)	Biological Reference Intervals	Unit(s)
Protein	Negative	Negative	
Ketones	Negative	Negative	
Bilirubins	Negative	Negative	
Nitrite	Negative	Negative	
Blood	Negative	Negative	
Urobilinogen	Negative	Negative	
Leukocytes	Negative	Negative	
MICROSCOPIC EXAMINATION			
Pus cells (WBCs)	3 - 4	0 - 2/hpf	/hpf
Epithelial cells	2 - 3	0 - 4/hpf	/hpf
Red blood cells	Nil	0 - 2/hpf	/hpf
Cast	Nil	Nil	
Crystals	Nil	Nil	
Other	Nil	Nil	

Note

URINE ALBUMIN/CREATININE RATIO

Test Description	Result(s)	Biological Reference Intervals	Unit(s)
*URINE ALBUMIN Latex Method(Sample Type:Urine)	5.14	0 - 15	mg/L
*URINE CREATININE Jaffe's Compensated(Sample Type:Urine)	45.06	37 - 250	mg/dL
*URINE ALBUMIN/CREATININE RATIO calculated	11.41	0 - 30	mg/gram creatinine

Instrument: Tests performed on Biosystems BA 400, A Fully Automatic Biochemistry analyser

Normal: 0 - 30
Microalbuminuria: 30 - 300 mg/gram creatinine
Macroalbuminuria : > 300 mg/gram creatinine

Note

E S R

Test Description	Result(s)	Biological Reference Intervals	Unit(s)
*ESR	7	0 - 15	mm/Hour
Method	Automated Westergren		

Interpretation:

- It indicates presence and intensity of an inflammatory process. It does not diagnose a specific disease. Changes in the ESR are more significant than the abnormal results of a single test.
- It is a prognostic test and used to monitor the course or response to treatment of diseases like tuberculosis, bacterial endocarditis, acute rheumatic fever, rheumatoid arthritis, SLE, Hodgkins disease, temporal arteritis and polymyalgia rheumatica.

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Test Description	Result(s)	Biological Reference Intervals	Unit(s)

- It is also increased in pregnancy, multiple myeloma, menstruation, and hypothyroidism.

Note

Electrolytes

Test Description	Result(s)	Biological Reference Intervals	Unit(s)
Sodium Method: ISE Direct(Sample Type:Serum)	137.2	136 - 145	mmol/L
Potassium Method: ISE Direct(Sample Type:Serum)	4.59	3.7 - 5.4	mmol/L
Ionised calcium Method: ISE Direct(Sample Type:Serum)	**5.41**	4.5 - 5.3	mg/dL
Chloride Method: ISE Direct(Sample Type:Serum)	**87.5**	90 - 110	mmol/L

Instrument:Test Done On Fully Automated Biochemistry Analyser, Sensa Core ST-200

Note

END OF REPORT

A. Kiran Kumar
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END OF REPORT

Approved By

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